

Optimizing Customer Retention with AI: A Value Proposition for Company A: Transforming Behavioral Data into Financial Stability

Transforming Churn Risk into Measurable
ROI for Company A

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High Customer Acquisition Costs (CAC) mandate a strategic shift toward maximizing Lifetime Value.



Unpredictable Churn erodes market share and severely disrupts revenue forecasting



The Opportunity: Leveraging historical telecom data to predict and preempt attrition before it occurs.

The Problem

Company A possesses vast, untapped historical telecommunication s data (100,000+ records) but suffers from reactive, rather than proactive, churn management.

The Objective

Identify the hidden operational and financial drivers of customer attrition to deploy targeted, cost-effective retention interventions.

The Hypothesis

Customers experiencing specific hardware age limits or billing frictions are highly predictable churn risks that can be saved with timely AI identification.

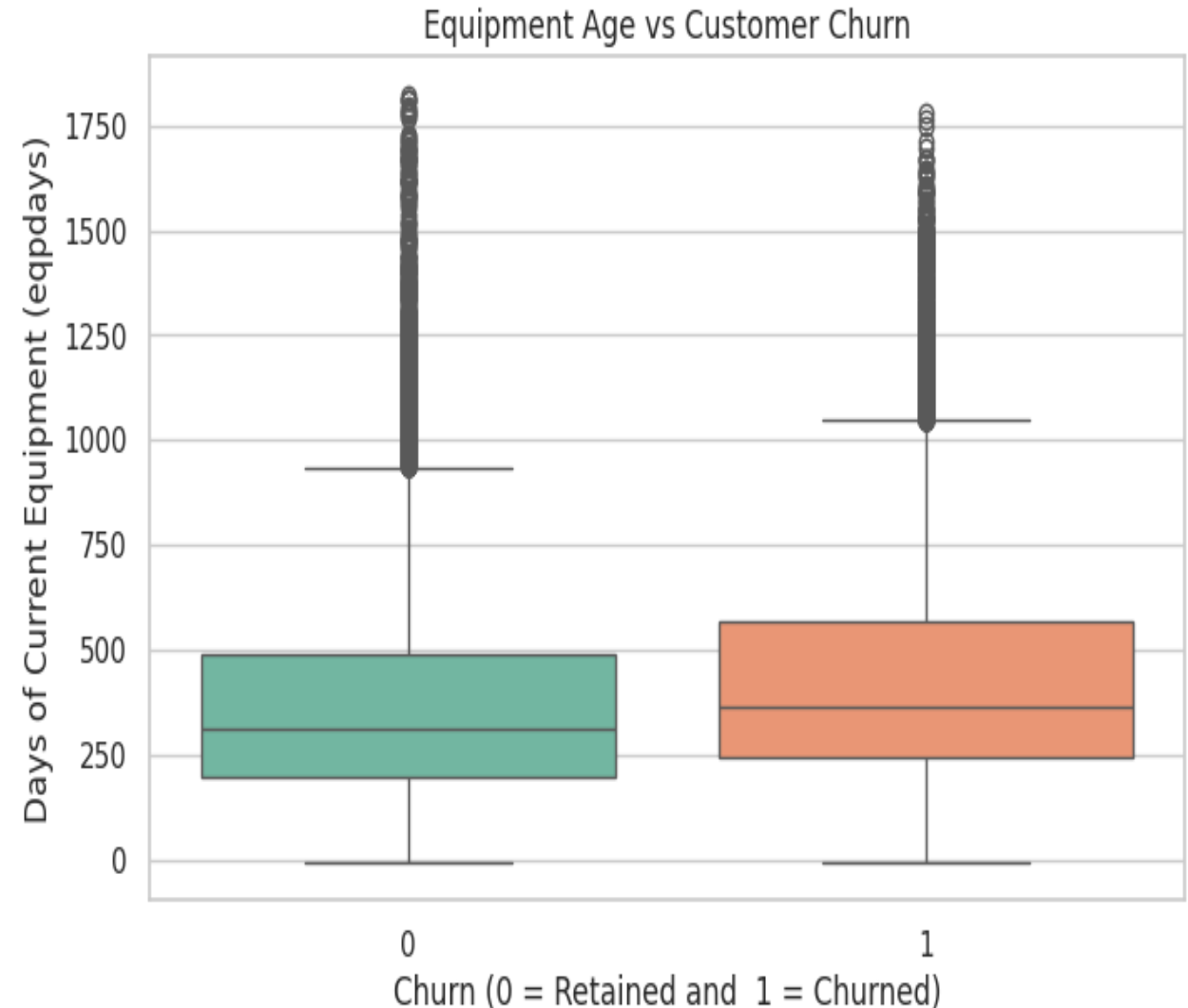
The Hardware Tipping Point

Key Insights

Customers operating with older handsets show a significantly higher propensity to leave.

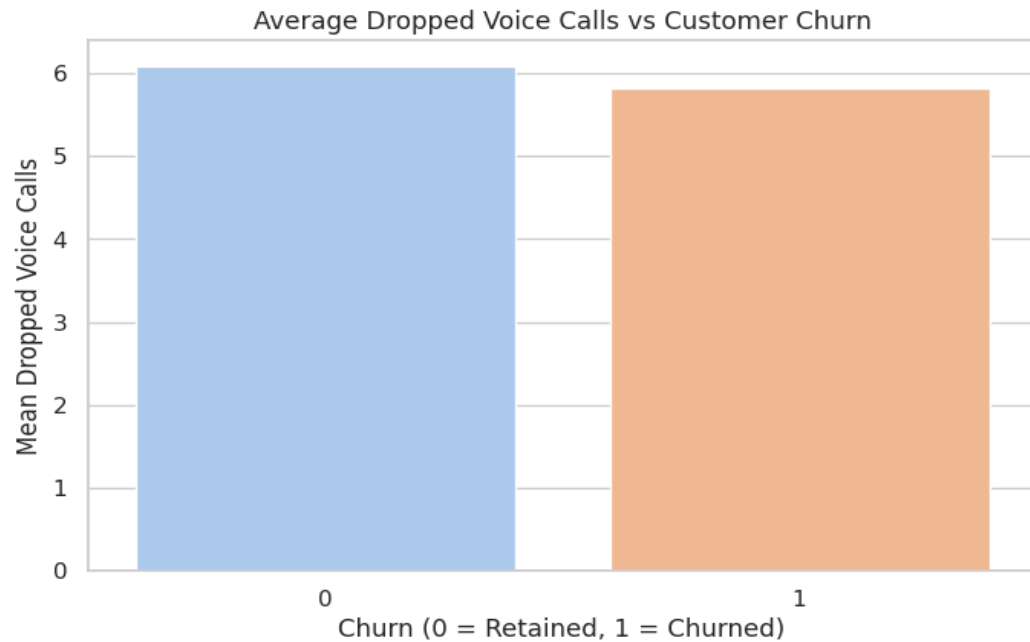
Hardware obsolescence is a primary, solvable trigger for attrition.

Upgrading equipment before this threshold is reached is critical for long-term retention



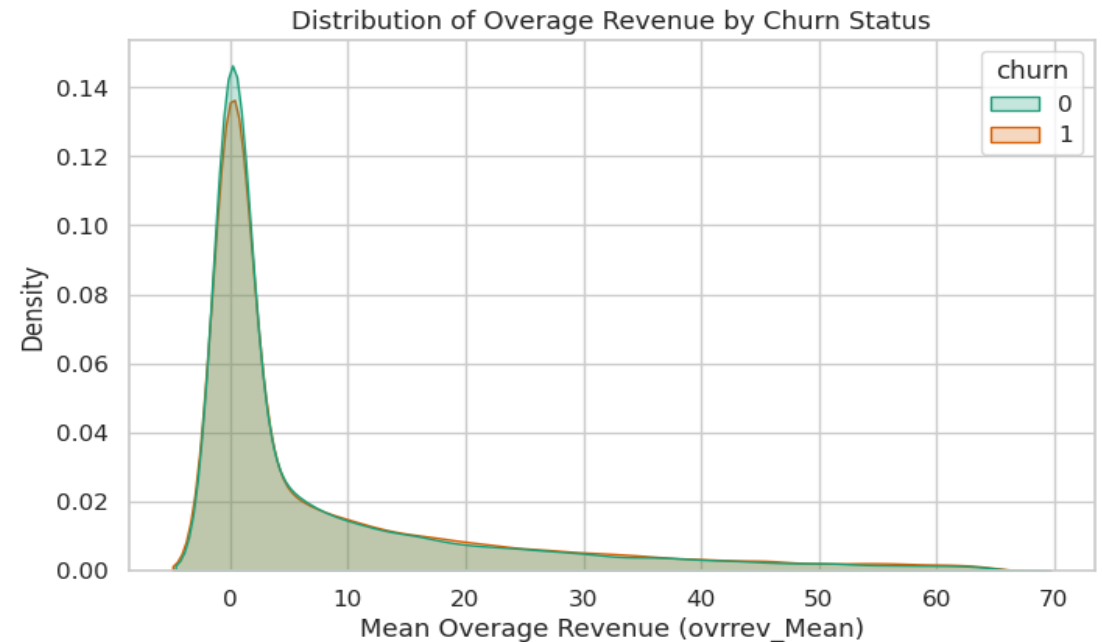
Dual threats: Network Reliability and Bill Shock

Service Quality



Retained vs. churned customers show measurably different tolerances for network failures and dropped voice calls (`drop_vce_Mean`).

Financial Friction

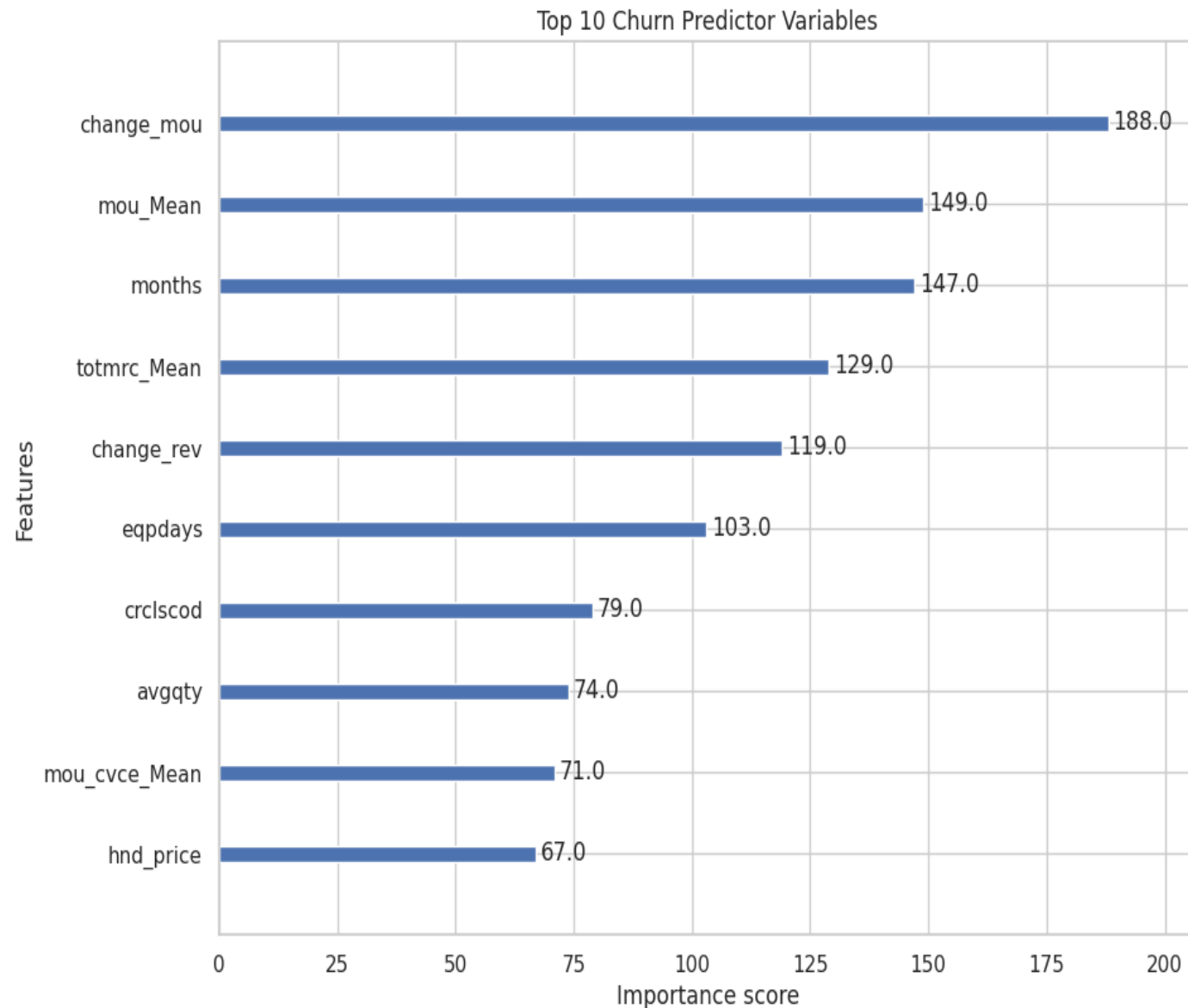


Churned customers exhibit a distinct, wider tail of overage charges (`ovrrev_Mean`), indicating that bill shock actively drives them to competitors.

The Algorithm's Verdict

Top Drivers

1. Sudden, massive percentage changes in monthly minutes of use (change_mou).
2. Total monthly recurring charges causing financial friction (totmrc_Mean).
3. Equipment age thresholds signaling hardware obsolescence (eqpdays).

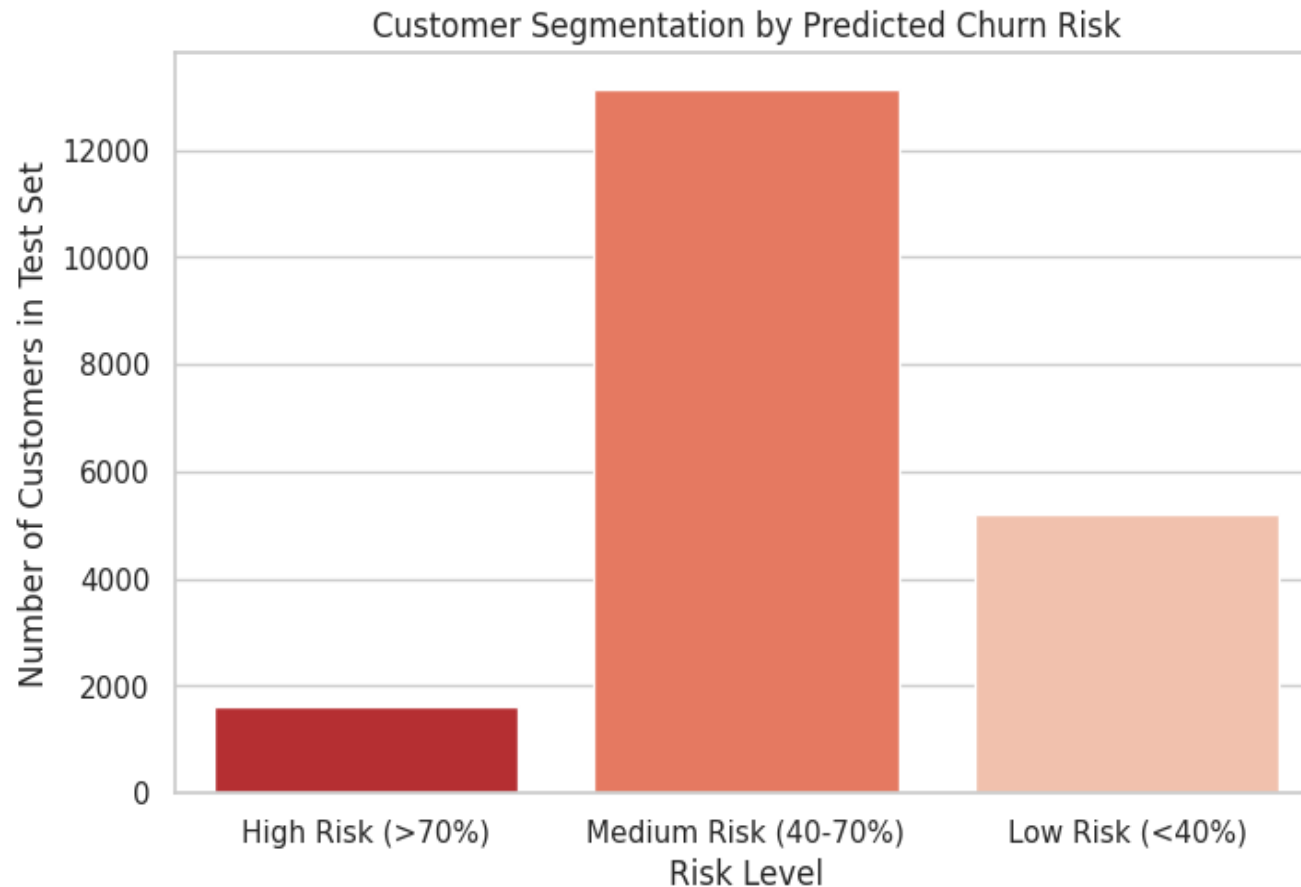


Model Evaluation Metric

AUC = 0.6891

Our predictive model significantly outperforms baseline guessing, providing a reliable, data-driven operational engine to identify high-risk accounts weeks before they actually churn.

Operationalizing the AI: Triage Strategy



High/Medium Risk (>40% Probability)

Trigger immediate proactive outreach. Offer targeted handset upgrades to neutralize hardware obsolescence, or tailored data plans to eliminate bill shock.

Low Risk (<40% Probability)

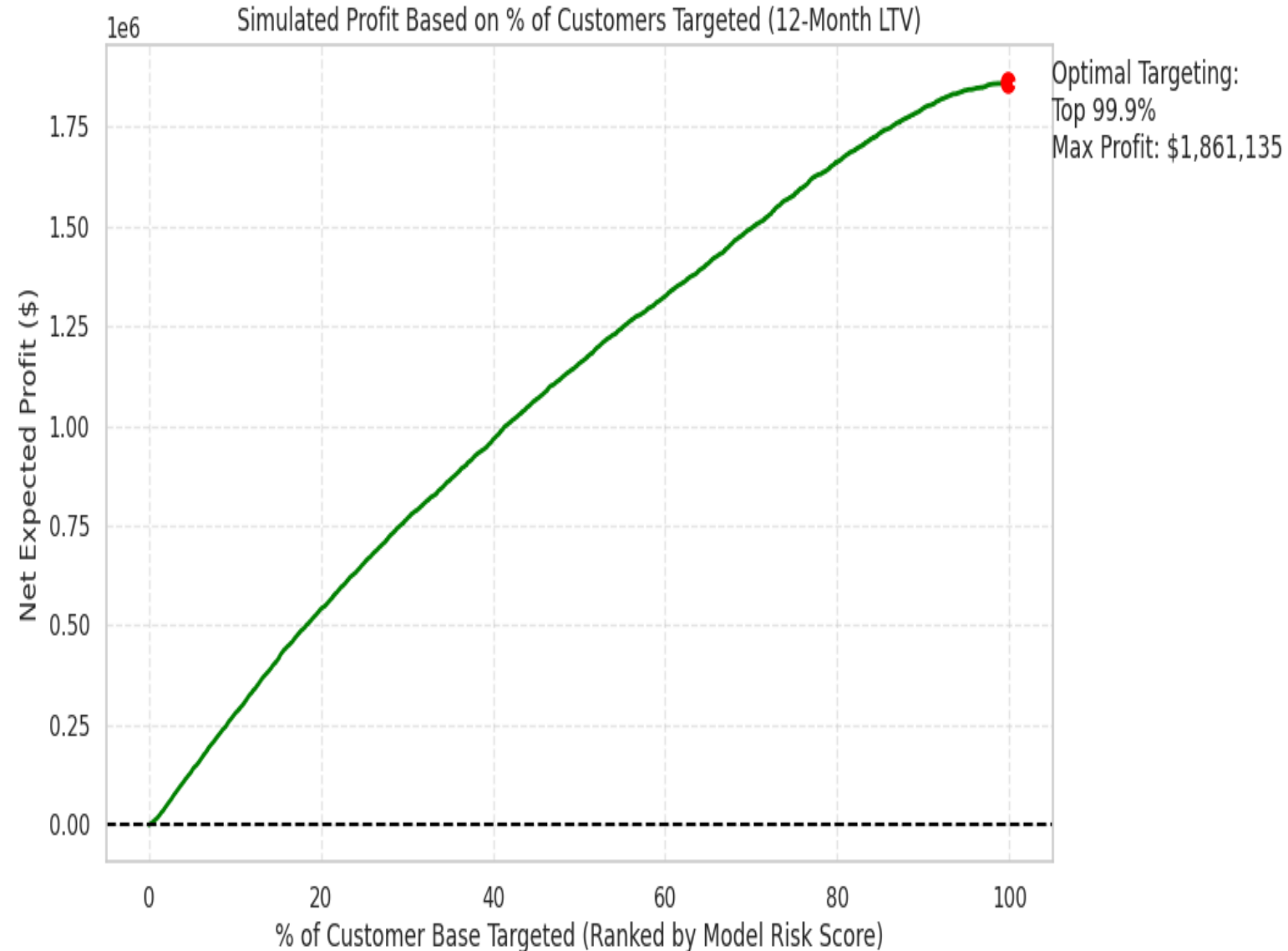
Maintain standard service protocol. Avoid unnecessary promotional spend to preserve profit margins.

ROI: The Financial Value of Retention

\$1,861,135

Maximum Projected 12 Month Net Profit.

*Based on optimal targeting of the top 99.9% at-risk base, accounting for intervention costs and lifetime customer value.





References

Company A telecom dataset (Omnicampus GCI World 2026).

Insights derived from Exploratory Data Analysis (pandas, seaborn) and Machine Learning modeling (XGBoost, scikit-learn).

AI link gemini: <https://gemini.google.com/share/aeb20979d7f7>

Google colab link: <https://colab.research.google.com/drive/1jZciWNnZxA-T4X4LYx74cUL8PW1vG6Jf?usp=sharing>